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ETHICAL CONSIDERATIONS

(1) **Intellectual property.** In our research, we utilized stories sourced from the Douban website and WritingPrompts. For the former, we strictly adhered to its copyright agreement. The latter is open-source, and we have complied fully with its licensing terms. We believe that all datasets have been properly desensitized and do not contain any personal information of the annotators or the original authors. We further adopt GPT-4o to check all human-written stories to flag potential sensitive content such as violence or explicit material. Actually, we find that there are no stories flagged as sensitive. We also randomly sample and check 10 human-written stories and find no sensitive content. For the Large Language Models (LLMs) investigated in this study, we queried LLMs (e.g., GPT-4o, Gemini 2.5 Pro, etc.) through their paid APIs. (2) **Intended Use.** This paper investigates the performance of reward models in the context of story generation and provides a high-quality benchmark and a reward model. Through the in-depth analysis presented, we aim to offer meaningful insights into LLMs and the task of story generation for the academic community. (3) **Misuse risks.** The reward model and benchmark presented in this paper are intended to improve story generation. No one should use our story generation method to create illegal stories, disseminate illicit content, or for commercial profit. (4) **AI assistance.** We used LLMs to generate data in our paper which have been stated in 2 and 3. We have used ChatGPT to refine some sentences. (5) **Worker Treatments** are discussed in Appendix C.2

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